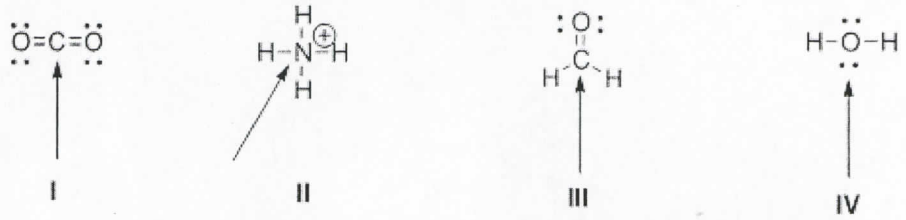
Use scantron for answers, not blue book!

Organic Cumulative Exam **Oct. 20, 2017**

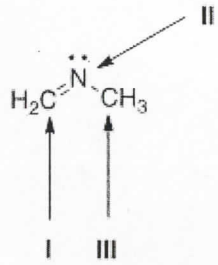
Multiple Choice. Choose the one alternative that best completes the statement or answers the question.

1) Determine the geometry around the indicated atom in each species. 1) _____



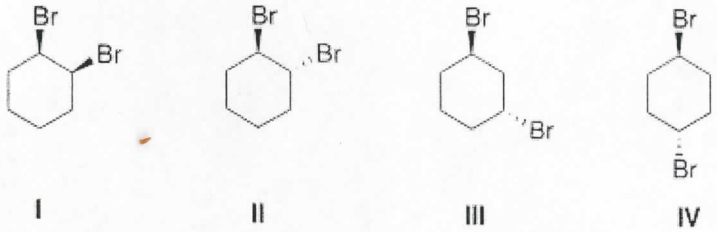
A) I = Linear; II = tetrahedral; III = trigonal planar; IV = linear
B) I = Tetrahedral; II = trigonal planar; III = linear; IV = tetrahedral
C) I = Trigonal planar; II = linear; III = tetrahedral; IV = trigonal planar
D) I = Linear; II = tetrahedral; III = trigonal planar; IV = tetrahedral

2) What is the hybridization for each of the indicated atoms in the following compound? 2) _____



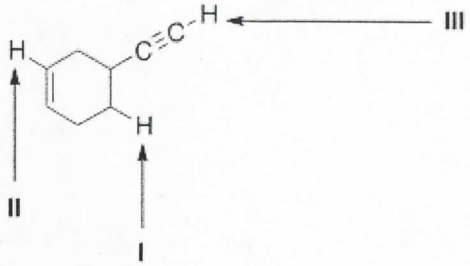
A) I = sp^2; II = sp^2; III = sp^3.
B) I = sp; II = sp^2; III = sp^3.
C) I = sp^2; II = sp^3; III = sp^3.
D) I = sp^2; II = sp^2; III = sp^2.

3) Which of the following molecules are chiral? 3) _____



A) I, IV
B) I, II
C) III, IV
D) II, III

4) Consider the following molecule with protons labeled, I-III. Rank these protons in order of decreasing acidity, putting the most acidic first. 4) _____



A) III > I > II B) I > III > II C) I > II > III D) III > II > I

5) Which of the following compounds is both a Brønsted-Lowry acid and base? 5) _____

CH_3OH CH_3COCH_3 H_2O $(CH_3)_3N$

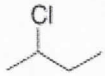
I II III IV

A) II, IV B) I, II C) I, IV D) I, III

6) What is the approximate C-C-C bond angle in propane? 6) _____

A) 180° B) 120° C) 109.5° D) 90°

7) Which of the following alkyl halides will react fastest with CH_3OH in an S_N1 mechanism? 7) _____



I

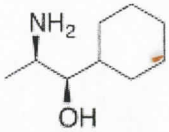
II

III

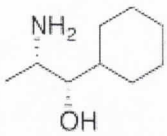
IV

A) I B) II C) III D) IV

8) What is the relationship between the following compounds? 8) _____



and



A) Enantiomers B) Constitutional isomers
C) Diastereomers D) Identical

9) Which of the following carbocations is the most stable? 9) _____

I II III IV

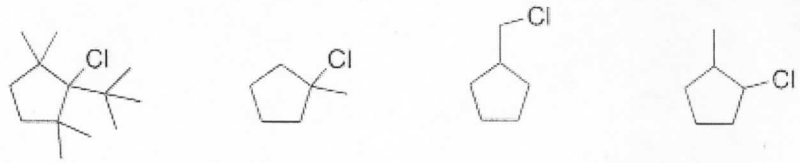
A) I B) II C) III D) IV

10) Which of the following compounds is (are) symmetrical ethers? 10) _____

I II III IV

A) Only III and IV
B) Only I
C) Only IV
D) Only I, III, and IV

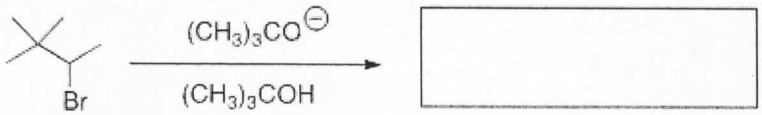
11) Which of the following is the least reactive substrate in an E2 reaction? 11) _____



I II III IV

A) I B) II C) III D) IV

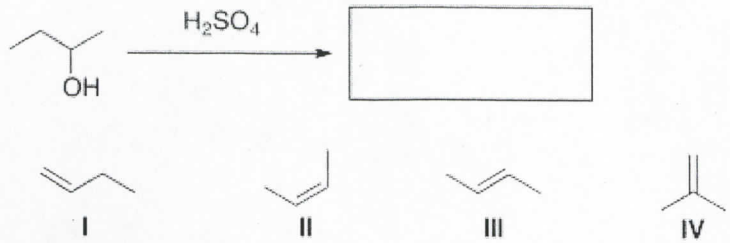
12) What is the major product of the following reaction? 12) _____



I II III IV

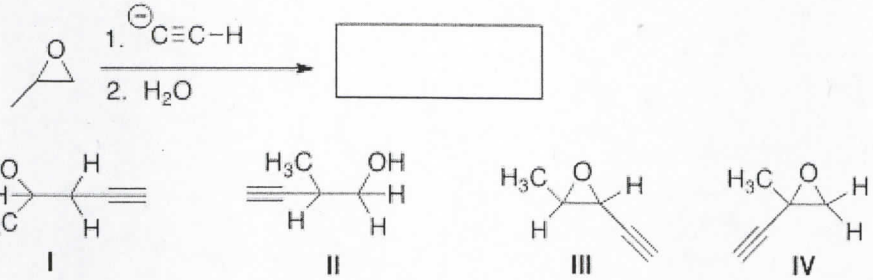
A) I B) II C) III D) IV

13) What is the major product of the following reaction? 13) _____



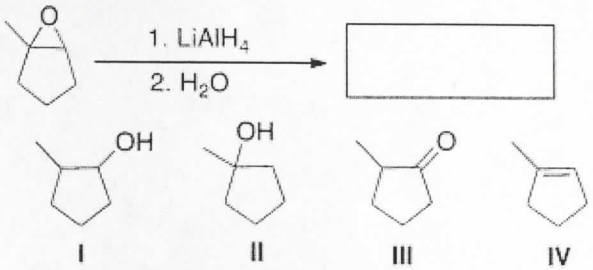
A) I B) II C) III D) IV

14) Determine the product of the following reaction. 14) _____



A) I B) II C) III D) IV

15) Determine the product of the following reaction. 15) _____

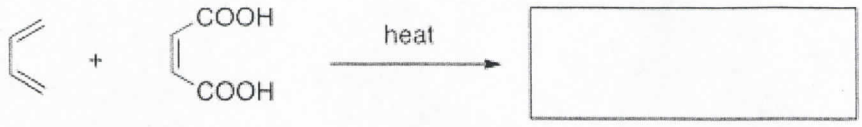


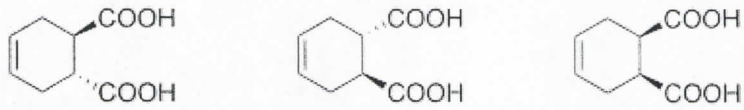
A) I B) II C) III D) IV

16) Which of the following ions is aromatic? 16) _____

A) I B) II C) III D) IV

17) What is the major organic product of the following reaction? 17) _____





I II III

A) Only II B) Only I and II C) Only I D) Only III

18) Arrange the following compounds in order of increasing boiling point, putting the compound with the lowest boiling point first. 18) _____

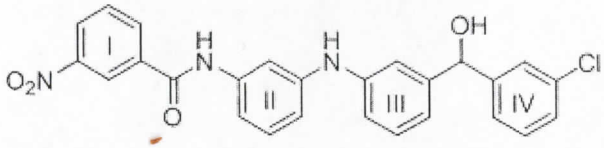
I II III IV

A) II < IV < I < III
B) I < II < III < IV
C) III < II < IV < I
D) I < IV < II < III

19) Which of the following reagents can accomplish the transformation below? 19) _____

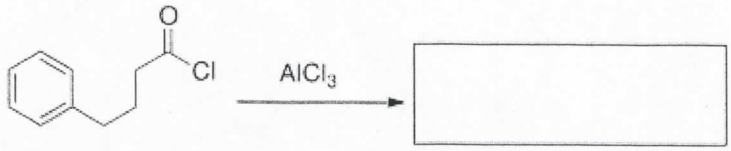
A) [1] O_3; [2] H_2O
B) [1] $LiAlH_4$, THF; [2] H_2O
C) $K_2Cr_2O_7$, H_2SO_4, H_2O
D) PCC, CH_2Cl_2

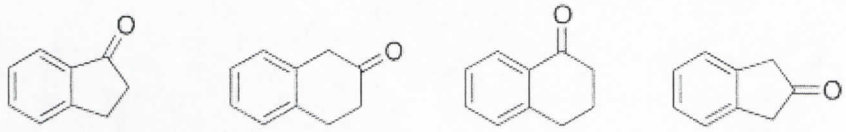
20) Consider the tetracyclic aromatic compound drawn below, with rings labelled as I, II, III, IV. Which of the four rings is least reactive in electrophilic aromatic substitution? 20) _____



A) I B) II C) III D) IV

21) What is the major organic product obtained from the following reaction? 21) _____

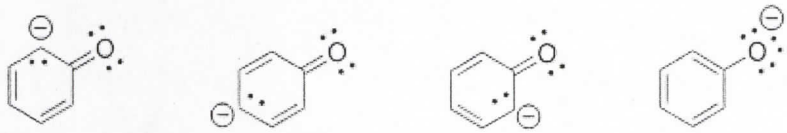




I II III IV

A) I B) II C) III D) IV

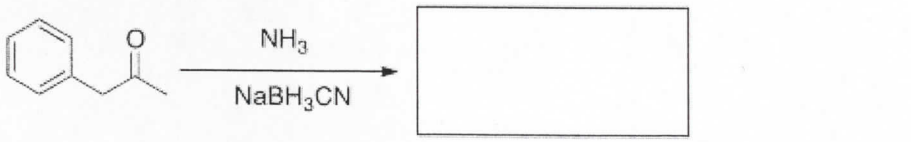
22) Which of the following structures is the major contributor to the resonance hybrid of the phenoxide anion? 22) _____

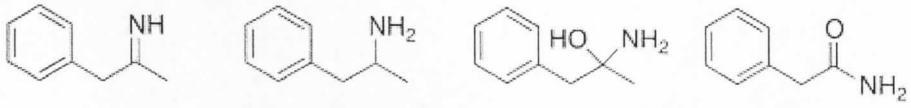


I II III IV

A) I B) II C) III D) IV

23) What is the major organic product obtained in the following reaction? 23) _____

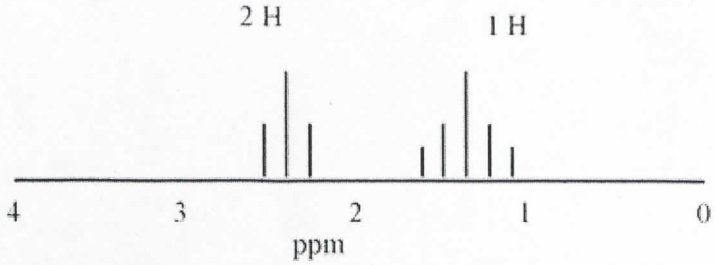




I II III IV

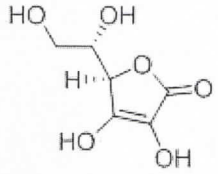
A) I B) II C) III D) IV

24) What compound is consistent with the following ^{1}H NMR spectrum? 24) _____

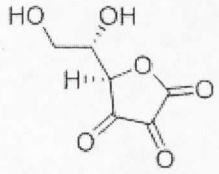


A) Cyclobutanone
B) 2-butanone
C) Acetone
D) Propanal

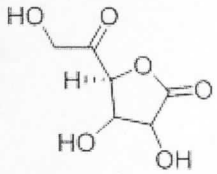
25) Vitamin C is a stable enediol. Which is the structure of a possible keto form in equilibrium with the enediol form? 25) _____



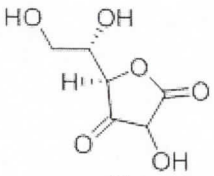
Vitamin C



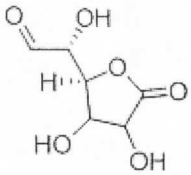
I



II



III



IV

A) I
B) II
C) III
D) IV